

Response to Reviewer Km2P:

Thank you for your valuable comments.

Weakness 1:

Our method is not a combination of DKD and CWD. The utilization of DKD is merely to explain the effectiveness of our method. Additionally, when the block size $K \times K$ is equal to the prediction size $H \times W$, our method degenerates into CWD. Thus, CWD is just a special case of our method under specific conditions, and our method can be seen as an extension of CWD.

(a) The differences between our method and DKD and CWD can be summarized as follows: Differences between our method and DKD:

Firstly, DKD is primarily designed for distillation in image classification and object detection, making it challenging to apply to semantic segmentation. Our method overcomes these challenges.

Secondly, we address the issue of pixel-wise segmentation map mismatch by dividing channels into sub-regions instead of relying on Ground Truth. DKD requires explicit target/non-target pixel labeling.

Thirdly, Our method focuses on local relationships between pixels within channels, while DKD emphasizes relationships between pixel categories.

Finally, We use the theory of DKD to explain our method's effectiveness but do not explicitly decouple target and non-target regions. The implicit decoupling in our method proves efficient and effective.

Differences between our method and CWD:

Firstly, CWD aims to improve distillation by aligning activation maps channel by channel, whereas our method believes that region-specific features within each channel contain more effective knowledge. Therefore, our approach uses a simple and direct block-wise partitioning to facilitate the transfer of region-specific feature knowledge.

Secondly, CWD calculates softmax over the entire channel, focusing on the overall inter-pixel relationships within the channel. In contrast, our RDKD calculates softmax within each region defined by channel partitioning, placing greater emphasis on the local relationships within channel-specific blocks.

(b) The differences between our method and Anchor-point Distillation [3] and HCL [4] are as follows:

Anchor-point Distillation uses simple average pooling, HCL employs a feature pyramid layer, while our method HRDKD performs multiple aggregations and calculates RDKD loss at each aggregation level.

Weakness 2:

Thank you for your valuable feedback. The comparative

experimental results with DKD and TaT are shown in the Table B in Official Comment.

Question 1: In the left part of line 217 in our paper, we provided the definitions of target regions and non-target regions. The definition stated: "Regions that encompass part or all target pixels are referred to as target regions, while regions that solely consist of non-target pixels are referred to as non-target regions." We do not explicitly differentiate between target and non-target blocks, even after aggregation operations like average pooling. We divide the model predictions into small blocks of size $K \times K$ across the entire channel to create local regions. Our goal is to align the student model with the teacher model on a local level.

Question 2:

There was a mismatch between the two formulas, and we intended to use the loss term LHRDKD. The corrected formula is now as follows.

$$L_{Total} = \eta L_{ce} + \mu L_{dice} + \gamma L_{HRDKD} \quad (1)$$

Limitations:

The limitations of our method include:

(1) In Table 3, there is no significant improvement in the mHD95 metric (related to boundary) when the teacher is TransUNet and the student is FCN-ERFNet.

(2) The hierarchical region fusion ability of HRDKD may require further improvement, especially on datasets like Pascal VOC.

(3) Our follow-up research is needed on cross-architecture knowledge transfer in natural image semantic segmentation, specifically between ViT Transformer and CNN architectures.